

OTEL-08: Troubleshooting OTel Pipelines

Series: OTEL — OpenTelemetry Integration | **Notebook:** 8 of 8 | **Created:** January 2026 | **Last Updated:** 08/25/2026

Debugging OpenTelemetry Data Flows

When telemetry data doesn't arrive as expected, systematic troubleshooting is essential. This notebook covers common issues, diagnostic techniques, and resolution steps for OpenTelemetry pipelines.

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Prerequisites

Requirement	Details
Knowledge	OTEL-01 through OTEL-07
Access	Collector logs, application logs

1. Troubleshooting Methodology

Systematic Approach

```
1. Identify the symptom
  ↓
2. Check component by component
   SDK → Collector → Backend
  ↓
3. Verify configuration
  ↓
4. Test connectivity
  ↓
5. Apply fix
  ↓
6. Verify resolution
```

Symptom Categories

Symptom	Likely Issue	Start Here
No data at all	Connectivity, auth	Section 4
Partial data	Filtering, sampling	Section 5
Delayed data	Performance	Section 6
Wrong data	Configuration	Section 5
Collector crashes	Resources	Section 6

2. Collector Issues

Collector Won't Start

Check configuration syntax:

```
# Validate config
otelcol-contrib validate --config=config.yaml
```

Common configuration errors:

Error	Cause	Fix
unknown type	Invalid component name	Check spelling, verify contrib has component
duplicate key	YAML syntax error	Check indentation
no pipelines defined	Missing service section	Add service.pipelines

Enable Debug Logging

```
service:
  telemetry:
    logs:
      level: debug
```

Collector Dropping Data

Check queue status:

```
# If using zpages extension
curl http://localhost:55679/debug/tracez
```

Common causes:

Cause	Indicator	Fix
Memory pressure	OOM, restarts	Increase memory_limiter limits

Cause	Indicator	Fix
Export failure	Retry logs	Check backend connectivity
Queue full	Queue overflow logs	Increase queue size or consumers

3. SDK Issues

No Spans Generated

Python:

```
# Enable SDK debug logging
import logging
logging.basicConfig(level=logging.DEBUG)

# Verify tracer provider is set
from opentelemetry import trace
print(trace.get_tracer_provider())
# Should NOT be NoOpTracerProvider
```

Java:

```
# Enable agent debug
java -javaagent:opentelemetry-javaagent.jar \
  -Dotel.javaagent.debug=true \
  -jar app.jar
```

Common SDK Issues

Issue	Cause	Fix
NoOpTracer	Provider not set	Call set_tracer_provider()
No export	Processor not added	Add BatchSpanProcessor
Missing spans	Not in context	Use start_as_current_span
Lost context	Async not handled	Use context propagation

Async Context Loss

```
# Wrong - loses context
async def bad_example():
    with tracer.start_as_current_span("parent"):
        await asyncio.gather(
            child_task(), # No context
            child_task()
        )

# Right - preserves context
async def good_example():
    with tracer.start_as_current_span("parent"):
        token = context.attach(context.get_current())
        try:
            await asyncio.gather(
                child_task(),
                child_task()
            )
        finally:
            context.detach(token)
```

4. Connectivity Issues

Test Collector Connectivity

```
# Test OTLP endpoint
curl -v http://collector:4318/v1/traces \
  -X POST \
  -H "Content-Type: application/json" \
  -d '{}'
```

```
# Test gRPC (with grpcurl)
grpcurl -plaintext collector:4317 list
```

Test Dynatrace Connectivity

```
# Test OTLP endpoint (set DT_API_TOKEN env var first)
curl -v https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces \
  -X POST \
  -H "Authorization: Api-Token ${DT_API_TOKEN}" \
  -H "Content-Type: application/json" \
  -d '{}'
```

Expected: 200 OK (empty body is fine)

Common Connectivity Errors

Error	Cause	Fix
connection refused	Wrong port/host	Verify endpoint
401 Unauthorized	Invalid token	Check token, regenerate
403 Forbidden	Missing scope	Add required scope to token
certificate error	TLS issue	Check certs, disable verify for test
timeout	Network/firewall	Check firewall rules

Proxy Issues

```
# Collector proxy configuration
exporters:
  otlphttp:
    endpoint: https://tenant.live.dynatrace.com/api/v2/otlp
    proxy_url: http://proxy.example.com:8080
```

```
# Or via environment
export HTTPS_PROXY=http://proxy.example.com:8080
```

5. Data Quality Issues

Missing Attributes

Debug exporter to see data:

```
exporters:
  debug:
    verbosity: detailed

service:
  pipelines:
    traces:
      exporters: [debug, otlphttp]
```

service.name Missing

Without `service.name`, Dynatrace can't create service entities:

```
# Add default service.name in collector
processors:
  resource:
    attributes:
      - key: service.name
        value: "unknown-service"
        action: insert # Only adds if missing
```

Wrong Time Zone

```
# Ensure collector uses UTC
# In Kubernetes:
env:
  - name: TZ
    value: UTC
```

Data Being Filtered

Check filter processor configuration:

```
processors:
  filter:
    traces:
      span:
        # This filters OUT matching spans
        - 'attributes["http.url"] == "/health"'
```

6. Performance Issues

Collector Memory Issues

Symptoms:

- OOMKilled restarts
- Slow processing
- Data drops

Fixes:

```
processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 1500      # Increase limit
    spike_limit_mib: 500 # Allow spikes

  batch:
    timeout: 5s          # Faster flush
    send_batch_size: 500 # Smaller batches
```

High Latency

Causes and fixes:

Cause	Indicator	Fix
Large batches	Delayed export	Reduce batch size
Slow export	Backend latency	Increase parallelism
Too many processors	Processing time	Remove unnecessary processors


```
exporters:
  otlphttp:
    sending_queue:
      num_consumers: 10 # More parallel exports
```

SDK Overhead

```
# Use sampling to reduce volume
from opentelemetry.sdk.trace.sampling import TraceIdRatioBased

provider = TracerProvider(
    sampler=TraceIdRatioBased(0.1), # 10% sampling
    resource=resource
)
```

7. Diagnostic Tools

Collector zpages

```
extensions:
  zpages:
    endpoint: 0.0.0.0:55679

service:
  extensions: [zpages]
```

Access:

- <http://collector:55679/debug/servicez> - Service info
- <http://collector:55679/debug/pipelinez> - Pipeline status
- <http://collector:55679/debug/extensionz> - Extensions
- <http://collector:55679/debug/tracez> - Trace samples

Health Check

```
extensions:
  health_check:
    endpoint: 0.0.0.0:13133
```

```
curl http://collector:13133/  
# {"status":"Server available"}
```

pprof Profiling

```
extensions:  
  pprof:  
    endpoint: 0.0.0.0:1777
```

```
go tool pprof http://collector:1777/debug/pprof/heap
```

```
// Check OTel data arrival  
fetch spans, from: now() - 1h  
| filter isNotNull(otel.scope.name)  
| summarize count = count(), by:{time_bucket = bin(timestamp, 5m)}  
| sort time_bucket asc
```

```
// Find services without service.name  
fetch spans, from: now() - 1h  
| filter isNull(service.name) or service.name == ""  
| summarize count = count(), by:{otel.scope.name}  
| sort count desc
```

8. Common Fixes

Quick Reference

Problem	Fix
No data at all	Check endpoint, auth token, network
No service entity	Add service.name resource attribute
Collector OOM	Increase memory_limiter, reduce batch size
High cardinality	Remove dynamic attributes
Context lost	Use context propagation in async

Problem	Fix
Wrong time	Set TZ=UTC
SSL errors	Check certs, add CA

Collector Config Template

A production-ready template:

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 1500
    spike_limit_mib: 500
  batch:
    timeout: 10s
    send_batch_size: 1000
  resource:
    attributes:
      - key: service.name
        value: unknown
        action: insert

exporters:
  otlphttp:
    endpoint: https://${DT_ENV}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_TOKEN}
    retry_on_failure:
      enabled: true
      initial_interval: 5s
      max_interval: 30s

extensions:
  health_check:
  zpages:

service:
  extensions: [health_check, zpages]
  telemetry:
    logs:
      level: info
  pipelines:
    traces:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp]
    metrics:
```

```
receivers: [otlp]
processors: [memory_limiter, resource, batch]
exporters: [otlphttp]
logs:
  receivers: [otlp]
  processors: [memory_limiter, resource, batch]
  exporters: [otlphttp]
```

Summary

In this notebook, you learned:

- Systematic troubleshooting methodology
- Collector debugging and configuration validation
- SDK issue diagnosis and async context handling
- Connectivity testing and common errors
- Data quality issue resolution
- Performance tuning for memory and latency
- Diagnostic tools: zpages, health check, pprof
- Quick fixes for common problems

References

- [Collector Troubleshooting](#)
- [Python instrumentation \(opentelemetry.io\)](#)
- [Dynatrace OTel Troubleshooting](#)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.